Beta & Gamma function

→ Beta function :- $(\beta(m,n)) = \beta(n,m)$

- $\beta(m,n) = \int_0^1 x^{m-1} \cdot (1-x)^{n-1}\, dx = \beta(n,m)$

property :-

(i) $\beta(m,n) = 2\int_0^{\pi/2} (\sin\theta)^{2m-1} (\cos\theta)^{2n-1}\, d\theta$

proof :-

$$\beta(m,n) = \int_0^1 x^{m-1} \cdot (1-x)^{n-1}\, dx$$

let $x = \sin^2\theta$

$$\frac{d}{dx} x = \frac{d}{dx} \sin^2\theta$$

$$1 = 2\sin\theta\cos\theta \frac{d\theta}{dx}$$

$$dx = 2\sin\theta\cos\theta \cdot d\theta$$

$$1 - x = 1 - \sin^2\theta$$

$$1 - x = \cos^2\theta$$

limit :-

x	0	1
θ	0	$\pi/2$

$$\int_0^{\pi/2} (\sin^2\theta)^{m-1} \cdot (\cos^2\theta)^{n-1} \cdot 2\sin\theta\cos\theta\, d\theta$$

$$2 \cdot \int_0^{\pi/2} (\sin\theta)^{2m-2} \cdot (\sin\theta)^1 \cdot (\cos\theta)^{2n-2} \cdot (\cos\theta)^1\, d\theta$$

$$\int_0^{\pi/2} (\sin\theta)^{2m-1} \cdot (\cos\theta)^{2n-1}\, d\theta = \frac{1}{2}\beta(m,n)$$

$$2\cdot\int_0^{\pi/2} (\sin\theta)^{2m-2+1} \cdot (\cos\theta)^{2n-2+1} \cdot d\theta$$

$$= 2\cdot\int_0^{\pi/2} (\sin\theta)^{2m-1} \cdot (\cos\theta)^{2n-1} \, d\theta$$

$$\boxed{\beta(m,n) = 2\cdot\int_0^{\pi/2} (\sin\theta)^{2m-1} \cdot (\cos\theta)^{2n-1} \, d\theta}$$

proved.

(ii) $\beta(m,n) = \beta(n,m)$

Proof:- $\beta(m,n) = \int_0^1 x^{m-1} \cdot (1-x)^{n-1} \, dx$

let $z = 1-x$

$dz = -dx$

$x = 1-z$

z	1	0
x	0	1

$$\int_1^0 (1-z)^{m-1} \cdot (z)^{n-1} \, -dz$$

$$-\int_1^0 z^{(n-1)} \cdot (1-z)^{m-1} \, dz$$

$$\int_a^b f(x) = -\int_b^a f(x)$$

$$\int_0^1 z^{n-1}(1-z)^{m-1}\,dz = \beta(n,m)$$

→ Gamma function :- $\Gamma(n)$

- $\Gamma(n) = \int_0^\infty e^{-x} \cdot x^{n-1}\,dx \quad , n>0$

- Properties :

(i) $\Gamma(1) = 1$

(ii) $\Gamma(2) = 1$

$\Gamma(3) = 2 = 2!$

$\Gamma(4) = 2\times3 = 3!$

$\Gamma(5) = 2\times3\times4 = 4!$

$\Gamma(n) = (n-1)!$

or

$\Gamma(n+1) = n!$

$\Gamma(0) = \infty$

(iii) $\Gamma(n) = (n-1)\,\Gamma(n-1)$

or

$\Gamma(n+1) = n\,\Gamma(n)$

$[n>1]$

Important

(iv) $\Gamma(1/2) = \sqrt{\pi}$

Important

(v) $\Gamma(n) = \dfrac{\Gamma(n+1)}{n} \quad [n<1]$

→ Relation between Beta and Gamma function:

(i) $$\beta(m,n) = \frac{\Gamma(m)\cdot\Gamma(n)}{\Gamma(m+n)}$$

(ii) $$\int_0^{\pi/2} \sin^p x \cdot \cos^q x \, dx = \frac{1}{2}\,\frac{\Gamma\left(\frac{p}{2}+\frac{1}{2}\right)\cdot\Gamma\left(\frac{q}{2}+\frac{1}{2}\right)}{\Gamma\left(\frac{p}{2}+\frac{q}{2}+1\right)}$$

proof:-

$$\cancel{\beta(m,n) = \int_0^1 \left(x^{m-1}\cdot(1-x)^{n-1}\right) dx}$$

$$\cancel{\text{let } z = \sin^2 x}$$

$$\cancel{dz = 2\sin x\cos x \, dx}$$

$$\beta(m,n) = 2\cdot\int_0^{\pi/2} (\sin x)^{2m-1} \cdot (\cos x)^{2n-1} \, dx$$

$$p = 2m-1$$

$$m = \frac{p+1}{2}$$

$$m = \frac{p}{2}+\frac{1}{2}$$

$$q = 2n-1$$

$$n = \frac{q}{2}+\frac{1}{2}$$

$$\beta\left(\frac{p}{2}+\frac{1}{2}, \frac{q}{2}+\frac{1}{2}\right) = \frac{\Gamma\left(p/2+1/2\right)\cdot\Gamma\left(q/2+1/2\right)}{\Gamma\left(\frac{p}{2}+\frac{1}{2}+\frac{q}{2}+\frac{1}{2}\right)}$$

$$\beta(m,n) = \frac{\Gamma\left(\frac{p}{2}+\frac{1}{2}\right)\cdot\Gamma\left(q/2+1/2\right)}{\Gamma\left(p/2+q/2+1\right)}$$

$$2\int_0^{\pi/2} (\sin x)^{2m-1}\cdot(\cos x)^{2n-1}\,dx = \frac{\Gamma\left(\frac{p+1}{2}\right)\cdot\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+1}{2}\right)}$$

$$\boxed{\int_0^{\pi/2} \sin^p x\cdot\cos^q x\,dx = \frac{1}{2}\cdot\frac{\Gamma\left(\frac{p+1}{2}\right)\cdot\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+1}{2}\right)}}$$

proved.

Question :-

1.) Evaluates :- (in both Beta and Gamma form)

(a) $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta$ (b) $\int_0^{\pi/2} \sqrt{\sec\theta}\, d\theta$ (c) $\int_0^{\pi/2} (\sqrt{\tan\theta} + \sqrt{\cot\theta})\, d\theta$

(d) $\int_0^{\pi/2} \sqrt{\sin\theta}\, d\theta \times \int_0^{\pi/2} \frac{1}{\sqrt{\sin\theta\, d\theta}}\, d\theta = \pi$ (Solve)

2.) $\int_0^1 x^m (1-x^n)^r\, dx$ [in gamma form]

3.) $\int_0^1 \frac{dx}{\sqrt{1-x^n}}$ [in gamma form]

4.) $\int_0^1 x^5 (1-x^2)^3\, dx$ [in both Beta and gamma form]

5.) Show that

(a) $\int_0^1 \left(\log\left[\frac{1}{x}\right]\right)^{n-1} dx = \Gamma(n)$

(b) $\int_0^{\pi/2} \sin^3 x \cdot \cos^{5/2} x\, dx = \frac{8}{77}$

(c) $\int_0^1 \frac{x^2\, dx}{\sqrt{1-x^4}} \times \int_0^1 \frac{dx}{\sqrt{1+x^4}} = \frac{\pi}{4\sqrt{2}}$

6. $\int_0^{\pi/2} \frac{1}{\sqrt{\sin 2\theta}}\, d\theta$ [in beta and gamma form]

7. Find:-

(a) $\Gamma\left(\frac{7}{2}\right)$ (b) $\Gamma(2.5)$ (c) $\beta(2.5, 1.5)$

(d) $\beta\left(\frac{8}{5}, \frac{3}{2}\right)$

8.) $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta$ [in gamma form]

9.) Solve:-

(a) $\int_0^{\infty} e^{-x^2} dx$ (b) $\int_0^{\infty} t^4 e^{-2t^2} dt$

(c) $\int_0^{\infty} x^m e^{-x^n} dx$ (d) $\int_0^{\infty} \sqrt{x} \cdot e^{-x^3} dx$

(e) $\int_0^{\pi/2} \left(\sqrt{\tan\theta} + \sqrt{\sec\theta}\right) d\theta$

(f) $\int_0^{1} x^3 (1-\sqrt{x})^5 dx$

10.) Show that:-

(a) $\int_0^{1} y^{q-1} \cdot \left(\log \frac{1}{y}\right)^{p-1} dy = \frac{\Gamma(p)}{q^p}$

(b) $\int_0^\infty \frac{x^{n-1}}{1+x}\,dx = \frac{\pi}{\sin n\pi}$

(c) $\int_0^\infty \frac{e^{-\sqrt{x}}}{x^{7/4}}\,dx = \frac{8\sqrt{\pi}}{3}$

(d) $\int_0^\infty \frac{x^c}{c^x}\,dx = \left(\frac{1}{\log c}\right)^{c+1} \cdot \Gamma(c+1)$

(e) $\Gamma(1/2) = \sqrt{\pi}$

Answer

1.) (a) $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta$

$$\int_0^{\pi/2} \frac{\sqrt{\sin\theta}}{\sqrt{\cos\theta}}\, d\theta$$

$$\int_0^{\pi/2} \frac{(\sin\theta)^{1/2}}{(\cos\theta)^{1/2}}\, d\theta$$

$$\int_0^{\pi/2} (\sin\theta)^{1/2} \cdot (\cos\theta)^{-1/2}\, d\theta$$

$$\cdot\ \beta(m,n) = 2\int_0^{\pi/2} (\sin\theta)^{2m-1} (\cos\theta)^{2n-1}\, d\theta$$

$2m-1 = 1/2$	$2n-1 = -1/2$
$2m = 1 + 1/2$	$2n = -1/2 + 1$
$m = \frac{3/2}{2}$	$2n = 1/2$
$m = 3/4$	$n = \frac{1/2}{2}$
	$n = 1/4$

$\frac{1}{2}\beta(3/4, 1/4)$ [in Beta form]

$$\beta(3/4, 1/4) = \frac{1}{2}\frac{\Gamma(3/4)\cdot\Gamma(1/4)}{\Gamma(3/4 + 1/4)}$$

or

(b)

$$\frac{1}{2}\frac{\Gamma\left(\frac{p}{2}+\frac{1}{2}\right)\cdot\Gamma\left(\frac{q}{2}+\frac{1}{2}\right)}{\Gamma\left(p/2+q/2+1\right)}$$

$$=\frac{\Gamma(3/4)\cdot\Gamma(1/4)}{2\cdot\Gamma(1)}$$

gamma form $=\dfrac{\Gamma(3/4)\cdot\Gamma(1/4)}{2}$ $\quad[\Gamma(1)=1]$

(b) $\displaystyle\int_0^{\pi/2}\sqrt{\sec\theta}\,d\theta$

$$\int_0^{\pi/2}\frac{1}{\sqrt{\cos\theta}}\,d\theta$$

$$\int_0^{\pi/2}\frac{1}{(\cos\theta)^{1/2}}\,d\theta$$

$$\int_0^{\pi/2}1\times\cos^{-1/2}\theta\,d\theta$$

$$\int_0^{\pi/2}\sin^0\theta\cdot\cos^{-1/2}\theta\,d\theta \qquad [\sin^0\theta=1]$$

$2m-1=0$	$2n-1=-1/2$
$m=1/2$	$n=1/4$

$\frac{1}{2}\cdot\beta(1/2,1/4)$ (Beta form)

$$\frac{\Gamma\left(\frac{0}{2}+1/2\right)\cdot\Gamma\left(\frac{-1/2}{2}+1/2\right)}{\Gamma\left(\frac{0}{2}+\frac{-1/2}{2}+1\right)}=\frac{\Gamma(1/2)\cdot\Gamma\left(-1/4+\frac{1}{2}\right)}{\Gamma(-1/4+1)}$$

$$=\frac{1}{2}\frac{\sqrt{\pi}\cdot\Gamma(1/4)}{\Gamma(3/4)}$$

(c) $\int_0^{\pi/2} \left(\sqrt{\tan\theta} + \sqrt{\cot\theta}\right) d\theta$

$$\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta + \int_0^{\pi/2} \sqrt{\cot\theta}\, d\theta$$

$$\underbrace{\int_0^{\pi/2} \frac{\sqrt{\sin\theta}}{\sqrt{\cos\theta}}\, d\theta}_{I_1} + \underbrace{\int_0^{\pi/2} \frac{\sqrt{\cos\theta}}{\sqrt{\sin\theta}}\, d\theta}_{I_2}$$

$I_1 = \int_0^{\pi/2} (\sin\theta)^{1/2} \cdot (\cos\theta)^{-1/2}\, d\theta \rightarrow$ Compare with Beta form.

$2m-1 = 1/2$	$2n-1 = -1/2$
$m = 3/4$	$n = 1/4$

$\beta(3/4, 1/4)$

$I_2 = \int_0^{\pi/2} (\cos\theta)^{1/2} \cdot (\sin\theta)^{-1/2}\, d\theta$

$n = 3/4, \quad m = 1/4$

$\beta(1/4, 3/4)$

$$\cdot \int_0^{\pi/2} \sin^p x \cdot \cos^q x \, dx = \frac{1}{2} \frac{\Gamma(p/2+1/2)\cdot\Gamma(q/2+1/2)}{\Gamma(p/2+q/2+1/2)}$$

$$\beta(3/4, 1/4) + \beta(1/4, 3/4) \quad \text{(Beta form)}$$

$$\frac{1}{2}\frac{\Gamma(1/4+1/2)\cdot\Gamma(-1/4+1/2)}{\Gamma(1/4-1/4+1)} + \frac{1}{2}\frac{\Gamma(-1/4+\frac{1}{2})\cdot\Gamma(1/4+1/2)}{\Gamma(-1/4+1/4+1)}$$

$$\frac{1}{2}\frac{\Gamma(3/4)\cdot\Gamma(1/4)}{\Gamma(1)} + \frac{\Gamma(1/4)\cdot\Gamma(3/4)}{\Gamma(1)}$$

$$\frac{1}{2}\, 2\,\Gamma(1/4)\cdot\Gamma(3/4) = \Gamma(1/4)\cdot\Gamma(3/4)$$

(d) $$\underbrace{\int_0^{\pi/2} \sqrt{\sin\theta}\, d\theta}_{I_1} \times \underbrace{\int_0^{\pi/2} \frac{1}{\sqrt{\sin\theta}}\, d\theta}_{I_2} = \pi$$

$$I_1 = \int_0^{\pi/2} (\sin\theta)^{1/2}\, d\theta$$

$$\int_0^{\pi/2} (\sin\theta)^{1/2} \cdot \cos^0\theta \, d\theta$$

$$\cdot \int_0^{\pi/2} \sin^p x \cdot \cos^q x \, dx = \frac{1}{2} \frac{\Gamma(p/2+1/2)\cdot\Gamma(q/2+1/2)}{\Gamma(p/2+q/2+1)}$$

$$\frac{1}{2}\,\frac{\Gamma\left(\frac{1/2}{2}+\frac{1}{2}\right)\cdot\Gamma\left(\frac{0}{2}+1/2\right)}{\Gamma\left(\frac{1/2}{2}+\frac{0}{2}+1\right)}$$

$$\frac{1}{2}\,\frac{\Gamma\left(1/4+1/2\right)\cdot\Gamma(1/2)}{\Gamma(1/4+1)}$$

$$\frac{1}{2}\,\frac{\Gamma(3/4)\cdot\sqrt{\pi}}{\Gamma(5/4)} \qquad \left(\Gamma(1/2)=\sqrt{\pi}\right)$$

$\cdot\ \Gamma(n)=(n-1)\Gamma(n-1) \quad [n>1]$ (property)

$$\frac{1}{2}\,\frac{\Gamma(3/4)\cdot\sqrt{\pi}}{\left(\frac{5}{4}-1\right)\Gamma(5/4-1)}$$

$$I_1=\frac{1}{2}\,\frac{\Gamma(3/4)}{\frac{1}{4}\cdot\Gamma(1/4)}\sqrt{\pi}$$

$$I_2=\int_0^{\pi/2}(\sin\theta)^{-1/2}\,d\theta$$

$$\int_0^{\pi/2} (\sin\theta)^{-1/2} \cdot \cos^0\theta \, d\theta$$

$$\frac{1}{2} \frac{\Gamma\left(\frac{-1/2}{2} + \frac{1}{2}\right) \cdot \Gamma\left(\frac{0}{2} + 1/2\right)}{\Gamma\left(\frac{-1/2}{2} + \frac{0}{2} + 1\right)}$$

$$\frac{1}{2} \frac{\Gamma(1/4) \cdot \Gamma(1/2)}{\Gamma(3/4)}$$

$$I_2 = \frac{1}{2} \frac{\Gamma(1/4) \cdot \sqrt{\pi}}{\Gamma(3/4)}$$

$$I_1 \times I_2$$

$$\frac{1}{2} \frac{\cancel{\Gamma(3/4)}}{1/4 \cdot \cancel{\Gamma(1/4)}} \sqrt{\pi} \times \frac{1}{2} \frac{\cancel{\Gamma(1/4)} \cdot \sqrt{\pi}}{\cancel{\Gamma(3/4)}}$$

$$\frac{1}{4} \times \frac{\sqrt{\pi} \cdot \sqrt{\pi}}{1/4} = \pi$$

2. $\int_0^1 x^m (1-x^n)^p \, dx$

let $z = x^n$, $x = z^{1/n}$

$dz = n x^{n-1} dx$

$dx = \dfrac{dz}{n x^{n-1}}$

$= \dfrac{dz}{n \cdot (z^{1/n})^{n-1}}$

x	0	1
z	0	1

$= \dfrac{dz}{n \cdot z^{\frac{n-1}{n}}}$

$\int_0^1 (z^{1/n})^m \cdot (1-z)^p \dfrac{dz}{n \cdot z^{\frac{(n-1)}{n}}}$

$\cancel{\frac{1}{n} \cdot \int_0^1 z^{m/n - \frac{n-1}{n}} \cdot (1-z)^p \, dz}$

$\cancel{\frac{1}{n} \cdot \int_0^1 z^{\frac{m+n+2}{n}} \cdot (1-z)^p \, dz}$

$\cancel{\int_0^1 x^{m-1} \cdot (1-x)^{n-1} \, dx}$

$\cancel{\beta(}$

$$\frac{1}{n}\int_0^1 z^{m/n} \cdot z^{-\left(\frac{n-1}{n}\right)} \cdot (1-z)^p \, dz$$

$$\frac{1}{n}\int_0^1 z^{\frac{m-n+1}{n}} \cdot (1-z)^p \, dz$$

$$\cdot\ \beta(M,N) = \int_0^1 x^{M-1} \cdot (1-x)^{N-1} \, dx$$

$M-1 = \frac{m-n+1}{n}$	$N-1 = p$
$M = \frac{m+1}{n}$	$N = p+1$

$$\beta(M,N) = \beta\left(\frac{m+1}{n}, p+1\right)$$

$$= \frac{\Gamma\left(\frac{m+1}{n}\right) \cdot \Gamma(p+1)}{\Gamma\left(\frac{m+1}{n} + p+1\right)}$$

3.) $\int_0^1 \frac{dx}{\sqrt{1-x^n}}$

let $z = x^n$ $\qquad x = z^{1/n}$

$dz = n x^{n-1} dx$

$$dx = \frac{dz}{n \cdot (z^{1/n})^{n-1}}$$

x	0	1
z	0	1

$$= \frac{dz}{n \cdot z^{n-1/n}}$$

$$\int_0^1 \frac{1}{\sqrt{1-z}} \frac{dz}{n \cdot z^{(n-1/n)}}$$

$$\frac{1}{n} \int_0^1 (1-z)^{-1/2} \cdot z^{\left(\frac{n-1}{n}\right)} \cdot dz$$

$$\beta(M,N) = \int_0^1 x^{M-1} (1-x)^{N-1} dx$$

$M = \frac{n-1}{n} + 1$ | $N = -1/2 + 1$

$M = \frac{2n-1}{n}$ | $N = 1/2$

$$\beta\left(\frac{2n-1}{n}, 1/2\right)$$

$$\frac{\Gamma\left(\frac{2n-1}{n}\right)\cdot\Gamma(1/2)}{\Gamma\left(\frac{2n-1}{n}+1/2\right)}$$

$$\frac{\Gamma\left(\frac{2n-1}{n}\right)}{\Gamma\left(\frac{4n-1}{2n}\right)}\cdot\sqrt{\pi}$$

4) $\int_0^1 x^5(1-x^2)^3\,dx$

let $z = x^2$, $x = \sqrt{z}$

$dz = 2x\,dx$ $\qquad x^5 = z^{5/2}$

$dx = \frac{dz}{2x}$

$= \frac{dz}{2\sqrt{z}}$

$$\int_0^1 z^{5/2}(1-z)^3\frac{dz}{2\sqrt{z}} = \frac{1}{2}\int_0^1 z^2(1-z)^3$$

$$\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$$

$m = 2+1$ | $n = 3+1$

$m = 3$ | $n = 4$

$\beta(3,4)$

$$\cancel{\beta(7/4, 4) = \frac{\Gamma(7/4)\cdot\Gamma(4)}{\Gamma(7/4+4)}}$$

$$\cancel{= \frac{\Gamma(7/4)\cdot 6}{\Gamma\left(\frac{23}{4}\right)}}$$

$\cancel{[\Gamma(4) = 3! = 6]}$

$$\beta(3,4) = \frac{\Gamma(3)\cdot\Gamma(4)}{\Gamma(3+4)}$$

$$= \frac{2!\cdot 3!}{6!}$$

$[\Gamma(n) = (n-1)!]$

$$= \frac{2\times 6}{6\times 5\times 4\times 3\times 2\times 1}$$

$$= \frac{1}{60}$$

5.) (a) $\int_0^1 \left(\log\left(\frac{1}{x}\right)\right)^{n-1} dx = \Gamma(n)$

$$LHS = \int_0^1 \left(\log\left(1/x\right)\right)^{n-1} dx$$

let $z = \log \frac{1}{x}$

$e^{z} = \frac{1}{x}$

$x = \frac{1}{e^{z}}$

$x = e^{-z}$

$dx = -e^{-z}\, dz$

x	1	0
z	0	∞

$$\int_{\infty}^{0} z^{n-1} \cdot -e^{-z}\, dz$$

$$-\int_{\infty}^{0} e^{-z} \cdot z^{n-1}\, dz$$

$$\int_{0}^{\infty} e^{-z} \cdot z^{n-1}\, dz.$$

$$\cdot\ \Gamma(n) = \int_{0}^{\infty} e^{-x} \cdot x^{n-1}\, dx$$

$$\therefore \int_{0}^{\infty} e^{-z} \cdot z^{n-1}\, dz = \Gamma(n)$$ proved.

(b) $\int_0^{\pi/2} \sin^3 x \cdot \cos^{5/2} x \, dx = \frac{8}{77}$

$$\int_0^{\pi/2} \sin^p x \cdot \cos^q x \, dx = \frac{1}{2} \frac{\Gamma\left(\frac{p}{2}+\frac{1}{2}\right) \Gamma\left(\frac{q}{2}+\frac{1}{2}\right)}{\Gamma\left(\frac{p}{2}+\frac{q}{2}+1\right)}$$

LHS:- $\frac{1}{2} \frac{\Gamma\left(\frac{3}{2}+\frac{1}{2}\right) \cdot \Gamma\left(\frac{5/2}{2}+\frac{1}{2}\right)}{\Gamma\left(\frac{3}{2}+\frac{5/2}{2}+1\right)}$

$$\frac{1}{2} \frac{\Gamma(2) \cdot \Gamma\left(\frac{5}{4}+\frac{1}{2}\right)}{\Gamma\left(\frac{3}{2}+1+\frac{5}{4}\right)}$$

$[\Gamma(2) = 1]$

$$\frac{\frac{1}{2} \cdot 1 \times \Gamma\left(\frac{7}{4}\right)}{\Gamma\left(\frac{6+2+5}{4}\right)}$$

$$\frac{1}{2} \cdot \frac{1 \cdot \Gamma\left(\frac{7}{4}\right)}{\Gamma\left(\frac{15}{4}\right)}$$

$$\frac{1}{2} \cdot \frac{\Gamma\left(\frac{7}{4}\right)}{\left(\frac{15}{4}-1\right) \Gamma\left(\frac{15}{4}-1\right)}$$

$[\Gamma(n) = (n-1)\Gamma(n-1)]$

$$\frac{\frac{1}{2}\cdot\Gamma(7/4)}{\frac{11}{4}\,\Gamma(11/4)}$$

$$\frac{\frac{1}{2}\cdot\Gamma(7/4)}{\frac{11}{4}\cdot(11/4-1)\cdot\Gamma(11/4-1)}$$

$$\frac{\frac{1}{2}\cdot\cancel{\Gamma(7/4)}}{\frac{11}{4}\times\frac{7}{4}\times\cancel{\Gamma(7/4)}}$$

$$=\frac{1}{2}\times\frac{1}{\frac{77}{16}}=\frac{16}{77\times 2}=\frac{8}{77}$$

(c) $\int_0^1 \frac{x^2\,dx}{\sqrt{1-x^4}} \times \int_0^1 \frac{dx}{\sqrt{1+x^4}} = \frac{\pi}{4\sqrt{2}}$

LHS:- $\underbrace{\int_0^1 \frac{x^2}{\sqrt{1-x^4}}\,dx}_{I_1} \times \underbrace{\int_0^1 \frac{dx}{\sqrt{1+x^4}}}_{I_2}$

$$I_1 = \int_0^1 \frac{x^2}{\sqrt{1-x^4}}\,dx$$

let $x^2 = \sin\theta$, $x = \sqrt{\sin\theta}$

$$2x\,dx = \cos\theta \cdot d\theta$$

$$dx = \frac{d\theta}{2x}\cos\theta$$

$$dx = \frac{d\theta}{2\sqrt{\sin\theta}\cdot\cos^{-1}\theta} =$$

x	0	1
θ	0	$\pi/2$

$$\int_0^{\pi/2} \frac{\sin\theta}{\sqrt{1-\sin^2\theta}} \cdot \frac{d\theta}{2\cos\theta\cdot\sqrt{\sin\theta}}$$

$$\frac{1}{2}\int_0^{\pi/2} \frac{\sin\theta}{\cos\theta\cdot\cos\theta\cdot\sqrt{\sin\theta}}\,d\theta$$

$$\frac{1}{2}\int_0^{\pi/2} \sin\theta^{1/2}\theta \; \cancel{(\cos\theta)^{-2}}\,d\theta$$

$$\frac{1}{2}\times\frac{1}{2}\,\frac{\Gamma\left(1/4+1/2\right)\cdot\Gamma\left(\frac{1}{2}+1/2\right)}{\Gamma\left(1/4+1\left(\frac{1}{2}\right)+1\right)}$$

$$\frac{1}{4}\,\frac{\Gamma(3/4)\cdot\Gamma(+1/2)}{\Gamma(5/4)}$$

$$I_2 = \int_0^1 \frac{dx}{\sqrt{1+x^4}}$$

let $x^2 = \tan\theta$ $\quad x = \sqrt{\tan\theta}$

$$2x\frac{dx}{d\theta} = \sec^2\theta$$

$$dx = \frac{\sec^2\theta\, d\theta}{2\sqrt{\tan\theta}}$$

x	0	1
θ	0	$\pi/4$

$$\int_0^{\pi/4} \frac{\sec^2\theta\, d\theta}{\sqrt{1+\tan^2\theta}\cdot 2\sqrt{\tan\theta}}$$

$$\frac{1}{2}\int_0^{\pi/4} \frac{\sec^2\theta\, d\theta}{\sec\theta\cdot\sqrt{\tan\theta}}$$

$$\frac{1}{2}\int_0^{\pi/4} \frac{\sec\theta}{\sqrt{\tan\theta}}\, d\theta$$

$$\frac{1}{2}\int_0^{\pi/4} \frac{\frac{1}{\cos\theta}}{\frac{\sqrt{\sin\theta}}{\sqrt{\cos\theta}}}\, d\theta$$

$$\frac{1}{2}\int_0^{\pi/4} \cos^{-1/2}\theta \cdot \sin^{-1/2}\theta \, d\theta$$

?

$$\frac{1}{2}\int_0^{\pi/4} \frac{1}{\sqrt{\sin\theta \cdot \cos\theta}}\, d\theta$$

multiply and divide $\sqrt{2}$

$$\frac{1}{2}\int_0^{\pi/4} \frac{\sqrt{2}}{\sqrt{2}\sqrt{\sin\theta\cos\theta}}\, d\theta$$

$$\frac{\sqrt{2}}{2}\int_0^{\pi/4} \frac{1}{\sqrt{2\sin\theta\cos\theta}}\, d\theta$$

$$\frac{1}{\sqrt{2}}\int_0^{\pi/4} \frac{1}{\sqrt{\sin 2\theta}}\, d\theta$$

let $y = 2\theta$

$dy = 2\,d\theta$

$d\theta = \frac{dy}{2}$

θ	0	$\pi/4$
y	0	$\pi/2$

$$\frac{1}{\sqrt{2}} \int_0^{\pi/2} \frac{1}{\sqrt{\sin y}} \frac{dy}{2}$$

$$\frac{1}{2\sqrt{2}} \int_0^{\pi/2} (\sin y)^{-1/2}\, dy$$

$$\frac{1}{2\sqrt{2}} \cdot \frac{1}{2} \frac{\Gamma(-1/4 + 1/2)\,\Gamma(0/2 + 1/2)}{\Gamma(-1/4 + 0 + 1)}$$

$$\frac{1}{4\sqrt{2}} \frac{\Gamma(1/4) \cdot \Gamma(1/2)}{\Gamma(3/4)}$$

$I_1 \times I_2$

$$\frac{1}{4} \frac{\Gamma(3/4) \cdot \Gamma(1/2)}{\Gamma(5/4)} \times \frac{1}{4\sqrt{2}} \frac{\Gamma(1/4)\,\Gamma(1/2)}{\Gamma(3/4)}$$

$$\frac{1}{4\sqrt{2}} \cdot \frac{\Gamma(1/2) \cdot \Gamma(1/2)\,\Gamma(1/4)}{1/4\,\Gamma(1/4)} = \frac{\pi}{4\sqrt{2}}$$

$[\Gamma(n) = (n-1)\Gamma(n-1)]$

6.) $$\int_0^{\pi/2} \frac{1}{\sqrt{\sin 2\theta}}\, d\theta$$

~~let $z = 2\theta$~~

~~$dz = 2\,d\theta$~~

~~$d\theta = \frac{dz}{2}$~~

$$\int_0^{\pi/2} \frac{1}{\sqrt{2\sin\theta\cos\theta}}\, d\theta$$

$$\frac{1}{\sqrt{2}} \int_0^{\pi/2} (\sin\theta)^{-1/2} (\cos\theta)^{-1/2}\, d\theta$$

$$\frac{1}{\sqrt{2}} \cdot \frac{1}{2} \frac{\Gamma\left(\frac{-1/2}{2} + 1/2\right) \cdot \Gamma\left(\frac{-1/2}{2} + 1/2\right)}{\Gamma\left(\frac{-1/2}{2} + \frac{-1/2}{2} + 1\right)}$$

$$\frac{1}{2\sqrt{2}} \frac{\Gamma(1/4) \cdot \Gamma(1/4)}{\Gamma(1/2)}$$

$$\frac{1}{2\sqrt{2}\pi} \Gamma(1/4) \cdot \Gamma(1/4)$$

1) (a) $\Gamma\left(\frac{7}{2}\right)$

- $\Gamma(n) = (n-1)\,\Gamma(n-1)$

$\left(\frac{7}{2}-1\right)\Gamma\left(7/2-1\right)$

$\frac{5}{2}\cdot\Gamma\left(5/2\right)$

$\frac{5}{2}\cdot\left(\frac{5}{2}-1\right)\Gamma\left(5/2-1\right)$

$\frac{5}{2}\cdot\frac{3}{2}\cdot\Gamma\left(3/2\right)$

$\frac{5}{2}\cdot\frac{3}{2}\cdot\left(\frac{3}{2}-1\right)\Gamma\left(3/2-1\right)$

$\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\;\Gamma\left(1/2\right)$

$\frac{15}{8}\sqrt{\pi}$

(b) $\Gamma(2.5)$

- $\Gamma(n) = (n-1)\,\Gamma(n-1)$

$\Gamma(2.5) = (2.5-1)\,\Gamma(2.5-1)$

$$\frac{16+15}{10}$$

$$(1.5)\cdot\Gamma(1.5)$$

$$(1.5)\ (1.5-1)\ \Gamma(1.5-1)$$

$$1.5\cdot 0.5\ \Gamma(0.5)$$

$$\frac{3}{2}\ \frac{15}{10}\times\frac{1}{2}\ \Gamma(1/2)$$

$$\frac{3}{4}\cdot\sqrt{\pi}$$

(c) $\beta(2.5, 1.5)$

$$\frac{\Gamma(2.5)\cdot\Gamma(1.5)}{\Gamma(2.5+1.5)}$$

$$\frac{\frac{3}{4}\sqrt{\pi}\cdot\frac{\sqrt{\pi}}{2}}{\Gamma(4)} = \frac{\frac{3\pi}{8}}{3!} = \frac{\pi}{16}$$

(d) $\beta\left(\frac{8}{5}, \frac{3}{2}\right)$

$$\frac{\Gamma\left(\frac{8}{5}\right)\ \Gamma(3/2)}{\Gamma\left(\frac{31}{10}\right)}$$

8.) $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta$

$\int_0^{\pi/2} (\sin\theta)^{1/2} \cdot (\cos\theta)^{-1/2}\, d\theta$

$$\frac{1}{2} \frac{\Gamma(1/4 + 1/2)\ \Gamma(-1/4 + 1/2)}{\Gamma(1/4 - \frac{1}{4} + 1)}$$

$$\frac{1}{2} \Gamma(3/4) \cdot \Gamma(1/4)$$

9.) (a) $\int_0^{\infty} e^{-x^2}\, dx$

let $z = x^2$, $x = \sqrt{z}$

$dz = 2x\, dx$

$dz = 2\sqrt{z}\, dx$

$dx = \frac{dz}{2\sqrt{z}}$

x	0	∞
z	0	∞

$$\int_0^{\infty} e^{-z} \cdot \frac{dz}{2\sqrt{z}}$$

$$\frac{1}{2} \int_0^{\infty} z^{1/2} \cdot e^{-z}\, dz$$

$$\cdot\ \Gamma(n) = \int_0^\infty e^{-x} \cdot x^{n-1}\, dx$$

$$(n-1) = 1/2$$

$$n = 3/2$$

$$\Gamma(3/2) = (3/2 - 1)!\,(3/2 - 1) = 1/2\sqrt{\pi}$$

(b) $\int_0^\infty t^4 e^{-2t^2}\, dt$

let $z = 2t^2$ $\qquad t = \frac{\sqrt{z}}{\sqrt{2}}$

$dz = 4t\, dt$

$dt = \frac{4\sqrt{2}}{\sqrt{z}}\, dz$

t	0	∞
z	0	∞

$$\int_0^\infty \frac{z^2}{4} \cdot e^{-z} \cdot \frac{4\sqrt{2}}{\sqrt{z}}\, dz$$

$$\sqrt{2} \int_0^\infty z^{3/2} \cdot e^{-z}\, dz$$

$$\cdot\ \Gamma(n) = \int_0^\infty e^{-x} \cdot x^{n-1}\, dx$$

$$n = 3/2 + 1$$

$$n = 5/2$$

$\sqrt{2} \cdot \Gamma(5/2)$

$\sqrt{2} \cdot \frac{3}{2} \times \frac{1}{2} \cdot \sqrt{\pi} = \frac{3\sqrt{\pi}}{2\sqrt{2}}$

(c) $\int_0^\infty x^m e^{-x^n} dx$

let $z = x^n$, $x = z^{1/n}$

$dz = n x^{n-1} dx$

$dx = \frac{dz}{n \cdot z^{\frac{n-1}{n}}}$

$\frac{1}{2}$	0	∞
	0	∞

$\int_0^\infty z^{\frac{m}{n}} \cdot e^{-z} \cdot \frac{dz}{n \cdot z^{\frac{n-1}{n}}}$

$\frac{1}{n} \int_0^\infty z^{\frac{m}{n} - \left(\frac{n-1}{n}\right)} \cdot e^{-z} dz$

$\frac{1}{n} \int_0^\infty e^{-z} \quad z^{\frac{m-n+1}{n}} dz$

$\Gamma(N) = \int_0^\infty e^{-x} \cdot x^{N-1} dx$

$N = \frac{m-n+1+n}{n}$

$N = \frac{m+1}{n}$

$\frac{1}{6} - \frac{2}{3}$ $\quad \frac{1-4}{6} = -\frac{3}{6} = -\frac{1}{2}$

$$\frac{1}{n}\,\Gamma\left(\frac{m+1}{n}\right)$$

(d) $\int_0^\infty \sqrt{x} \cdot e^{-x^3}\,dx$

let $z = x^3$, $x = z^{1/3}$

$dz = 3x^2\,dx$ $\quad x^2 = z^{2/3}$

$\sqrt{x} = x^{1/6}$

$dx = \frac{dz}{3 \cdot z^{2/3}}$

$$\int_0^\infty z^{1/6} \cdot e^{-z} \cdot \frac{dz}{3 \cdot z^{2/3}}$$

$$\int_0^\infty e^{-z} \cdot z^{-1/2}\,dz$$

- $\Gamma(n) = \int_0^\infty e^{-z} \cdot z^{n-1}\,dz$

$n - 1 = -1/2$

$n = 1/2$

$\Gamma(1/2) = \sqrt{\pi}$

c) $\displaystyle\int_0^{\pi/2} \sqrt{\tan\theta} + \sqrt{\sec\theta}\ d\theta$

$$\underbrace{\int_0^{\pi/2} \frac{\sqrt{\sin\theta}}{\sqrt{\cos\theta}}\, d\theta}_{I_1} + \underbrace{\int_0^{\pi/2} \frac{1}{\sqrt{\cos\theta}}\, d\theta}_{I_2}$$

$$I_1 = \int_0^{\pi/2} (\sin\theta)^{1/2} (\cos\theta)^{-1/2}\, d\theta$$

$$\frac{1}{2}\, \frac{\Gamma(1/4 + 1/2)\,\Gamma(-1/4 + 1/2)}{\Gamma(1/4 - 1/4 + 1)}$$

$$\frac{1}{2}\, \Gamma(3/4) \cdot \Gamma(1/4)$$

$$I_2 = \frac{1}{2}\, \frac{\Gamma(0 + 1/2)\,\Gamma(-1/4 + 1/2)}{\Gamma(0 + 1 - 1/4)}$$

$$I = \frac{1}{2} \cdot \frac{\Gamma(1/2) \cdot \Gamma(1/4)}{\Gamma(3/4)}$$

$$\frac{1}{4}\, \Gamma(1/4) \cdot \Gamma(1/4) \cdot \sqrt{\pi}$$

(f) $\int_0^1 x^3(1-\sqrt{x})^5\,dx$

let $z=\sqrt{x}$, $x=z^2$

$dz=\frac{1}{2\sqrt{x}}\,dx$

$dx = 2z\,dz$

x	0	1
z	0	1

$\int_0^1 z^6(1-z)^5\,dz\cdot 2z$

$\frac{1}{2}\int_0^1 z^7(1-z)^5\,dz$

$\beta(m,n)=\int_0^1 x^{m-1}\cdot(1-x)^{n-1}\,dx$

$m-1=7$

$m=8 \quad n=6$

$\beta(8,6)=\frac{\Gamma(8)\cdot\Gamma(6)}{\Gamma(14)}=\frac{7!\times 5!}{13\times12\times11\times10\times9\times8\times7}$

(a)(q) $\int_0^1 y^{q-1}\left(\log\frac{1}{y}\right)^{p-1} dy = \frac{\Gamma(p)}{q^p}$

LHS:- let $z = \log\left(\frac{1}{y}\right)$

$$e^z = \frac{1}{y}$$

$$y = e^{-z}$$

$$dy = -e^{-z}\cdot dz$$

$$\int_\infty^0 (e^{-z})^{q-1}\cdot(z)^{p-1}\cdot -e^{-z}\cdot dz$$

y	0	1
z	∞	0

$$-\int_\infty^0 (e^{-z})^{q-1}\cdot z^{p-1}\cdot dz\, e^{-z}$$

$$\int_0^\infty (e^{-z})^{q}\cdot \cancel{(e^{-z})^{-1}}\,\cancel{(e^{-z})^{1}} + z^{p-1} dz$$

$$\int_0^\infty (e^{-z})^{q}\cdot z^{p-1} dz$$

let $t = zq$ $\qquad z = \begin{cases} 0 & t \\ \infty & q \end{cases}$

$$dt = q\,dz$$

$$dz = \frac{dt}{q}$$

$$\int_0^\infty e^{-t} \cdot \left(\frac{t}{q}\right)^{p-1} \frac{dt}{q}$$

$$\int_0^\infty e^{-t}\, t^{p-1} \cdot \frac{1}{q^{p-1} \cdot q^1}$$

$$\frac{1}{q^p} \int_0^\infty e^{-t}\, t^{p-1}\, dt$$

$$n = p-1+1$$

$$n = p$$

$$\frac{1}{q^p}\, \Gamma(p)$$

(b) $\int_0^\infty \frac{x^{n-1}}{1+x}\, dx = \frac{\pi}{\sin n\pi}$

let $y = \frac{1}{1+x}$

$$1+x = \frac{1}{y}$$

$$x = \frac{1}{y} - 1$$

$$dx = -\frac{1}{y^2}\, dy$$

x	0	∞
y	1	0

$$\int_1^0 \left(\frac{1}{y}-1\right)^{n-1} \cdot x \cdot \frac{-1}{y^2}\, dy$$

$$= \int_1^0 \frac{(1-y)^{n-1}}{y^{n-1}} \cdot \frac{1}{y}\, dy$$

$$+ \int_0^1 (1-y)^{n-1}\, y^{-n}\, dy$$

$$\cdot\ \beta(M,N) = \int_0^1 x^{M-1} (1-x)^{N-1}\, dx$$

$M-1=-n$ | $N=n$

$M=1-n$

$$\beta((1-n), n) = \frac{\Gamma(1-n)\cdot\Gamma(n)}{\Gamma(1-n+n)}$$

$$\Gamma(n)\cdot\Gamma(1-n)$$

~~let $z = 1+x$~~
~~$dz = dx$~~

x		
z		

(b) $\int_0^\infty \frac{x^{n-1}}{1+x}\,dx$?

(c) $\int_0^\infty \frac{e^{-\sqrt{x}}}{x^{7/4}}\,dx = \frac{8\pi}{\sqrt{3}}$

let $z = \sqrt{x}$ $\quad x = z^2$

$dz = \frac{1}{2\sqrt{x}}\,dx$

x	0	∞
z	0	∞

$dx = 2z \cdot dz$

$\int_0^\infty \frac{e^{-z}}{z^{7/2}}\, 2 \cdot z\, dz$

$2\int_0^\infty e^{-z}\, z^{-5/2}\, dz$

~~$2 \cdot \Gamma(-5/2)$~~ $\quad \left[\Gamma(n) = \frac{\Gamma(n+1)}{n}\right]$

~~$2 \cdot \frac{\Gamma(-3/2)}{-5}$~~ $\quad n<1$

$$2 \cdot \frac{\Gamma(-1/2)}{-\frac{5}{2} \cdot -\frac{3}{2}}$$

$$2 \times \frac{\Gamma(1/2)}{-\frac{5}{2} \times -\frac{3}{2} \times -\frac{1}{2}}$$

$$2$$

$$\Gamma(n) = \int_0^\infty e^{-x} \cdot x^{n-1}\, dx$$

$$n-1 = -5/2$$

$$n = -3/2$$

$$2 \cdot \Gamma(-3/2)$$

$$\left[\Gamma(n) = \frac{\Gamma(n+1)}{n} \quad (n<1) \right]$$

$$2 \cdot \frac{\Gamma\left(-\frac{3}{2}+1\right)}{-3/2}$$

$$2 \times \frac{\Gamma(-1/2)}{-3/2}$$

$$2 \times \frac{\Gamma(1/2)}{-3/2 \times \frac{1}{2}} = \frac{8\sqrt{\pi}}{3}$$

(d) $\int_0^\infty \frac{x^c}{c^x} dx = \left(\frac{1}{\log c}\right)^{c+1} \cdot \Gamma(c+1)$

let $z = c^x$

$\log z = \log c^x$

$\log z = x \log c$, $x = \frac{\log z}{\log c}$

$e^{\log z} = e^{x \log c}$

$z = e^{x \log c}$

$c^x = e^{x \log c}$

$\int_0^\infty \frac{x^c}{e^{x \log c}} \cdot dx$

$\int_0^\infty e^{-x \log} \cdot x^c \cdot dx$

let $y = x \log c$ $\quad x = \frac{y}{\log c}$

$dy = \log c \, dx$

$dx = \frac{dy}{\log}$

$\int_0^\infty e^{-y} \cdot \frac{y^c}{(\log c)^c} \cdot \frac{dy}{\log c}$

$$\frac{1}{(\log c)^{c+1}} \int_0^\infty e^{-y} \cdot y^c \, dy$$

$$\Gamma(n) = \int_0^\infty e^{-x} \cdot x^{n-1} \, dx$$

$n = c+1$

$$\frac{1}{(\log c)^{c+1}} \Gamma(c+1)$$ — proved..

$$\Gamma(1/2) = \sqrt{\pi}$$

Not mandatory